

# Prediction of Germination Ability of Tomato Seeds Based on Phenotypic Traits using Machine Learning

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**Abstract**— This research proposes a method to automatically detect the phenotypic traits of tomato seeds and predict the germinating ability before planting. The proposed autonomous screening method combines computer vision, and machine learning algorithms to provide real-time, non-invasive phenotypic analyses of tomato seeds. Seeds of tomato variety called “Thilina” which is one of the popular commercially grown tomato varieties in Sri Lanka were used for this experiment. An experimental setup with a camera was used for the acquisition of images of the seeds. The images were pre-processed and morphological traits were collected in every seed. Different machine learning models are trained and tested. Identification of the most effective model for predicting seed germination is determined based on the evaluation parameters of precision, recall, F1-score and accuracy. It is concluded that KNN is the best classifier where 94% of the test dataset is accurately predicted by the KNN model. The highlights of this study will be crucial to overcoming the labor-intensive nature and errors made when conducting conventional germination tests. To the best of our knowledge, this study is the first in Sri Lanka which automated conventional seed tests to predict the germination ability.

**Keywords**—phenotyping, computer vision, machine learning, tomato, seed germination

## I. INTRODUCTION

Morphological phenotyping of tomato seeds involves the study and classification of physical characteristics exhibited by the seeds of tomato plants. Morphological phenotyping of tomato seeds can provide valuable information for plant breeders, researchers, and growers. It can aid in selecting seeds with desired traits, such as size, shape, and color, as well as those that exhibit better germination rates and overall seed quality. This information is important for developing new tomato varieties that are better suited for specific growing conditions, culinary purposes, or market preferences. The morphological traits of tomato seeds include size, shape, color, surface texture, and distinguishing features such as markings or irregularities [1]. There are some of the morphological traits that can be observed and analyzed in Tomato seeds: The *size* of the seed may influence its nutrient reserves and initial growth rate. The *shape* of the Tomato

seeds is generally flat and oval in shape, with one side being slightly concave and the other slightly convex. The *texture* of the seed coat may influence water absorption during germination, and the *thickness* of the seed coat can affect the seed's ability to absorb water and nutrients.

Inconsistency in data, subjectivity and bias due to human interactions and inaccuracies are common problems in conventional phenotyping. Therefore, breeders need a transition from traditional methods to autonomous screening approaches. By automating the phenotyping process, the system significantly reduced labor costs and increased throughput, making it a cost-effective solution.

Machine learning (ML) can be considered as the ability of computers to learn without being explicitly programmed. Machine learning algorithms, inspired by the human learning process, iteratively learn from data and allow computers to find hidden insights. In order to find the most effective method for accurately evaluating the germination ability of tomato seeds, five classification methods such as K-Nearest Neighbors (KNN), Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), Decision Trees, and Deep Neural Networks (DNNs) were implemented and compared in this research.

CNN is a type of neural network that is particularly well-suited for image classification tasks. They can learn hierarchical patterns in image data, which makes them very effective at identifying and classifying different plant features. KNNs is a simple yet powerful algorithm that is often used for classification and regression tasks. It works by finding the K nearest neighbors of a new data point in the training set and then predicting the label or value of the new data point based on the labels or values of its neighbors. KNN is a non-parametric algorithm, which means that it does not make any assumptions about the underlying data distribution. This makes it a versatile algorithm that can be used with a wide variety of data types. SVM is very effective at finding non-linear decision boundaries, which makes them well-suited for tasks where the classes are not linearly separable. DTrees is a type of tree-based learning algorithm that is often used for classification and regression tasks. They work by

recursively splitting the data into smaller and smaller subsets until a stopping criterion is met. Decision trees are easy to interpret and can be very effective at modeling complex relationships between variables. DNN is a type of neural network that has multiple hidden layers.

Different imaging techniques have been adopted throughout the literature for the non-invasive characterization of tomato seeds. Galletti *et al.* [2] focus on the development of light-based methods for seed quality prediction in the agricultural industry. Optical imaging techniques, specifically chlorophyll fluorescence and chemometric-based multispectral imaging, were employed. The study involved tomatoes and carrots as test plants and compared these optical imaging techniques to traditional analytical methods for seed quality assessment. The results demonstrated that chlorophyll fluorescence-based technology can effectively discriminate cultivars and identify seed lots with lower physiological potential. Multispectral imaging, combined with a Random Forest classifier and quadratic discrimination analysis, allowed for accurate classification of seed lots with high- and low-vigor seeds. The analysis also revealed that low-quality seeds exhibit altered photosynthetic capacity and chlorophyll content. Regina *et al.* [3] used X-ray image analysis ray (8-bit) Image to evaluate the internal morphology of tomato hybrid seeds at different developmental stages. The results showed that seeds with normal embryos had higher vigor and produced normal seedlings. Seeds with larger internal free areas had lower vigor and a higher percentage of abnormal seedlings. The X-ray analysis proved to be an efficient method for studying seed physiological maturity and predicting seed performance based on embryo morphology. Gargiulo *et al.* [4] X-ray microtomography was used to investigate the relationship between seed physical characteristics and quality. The study found that the 3D morphology of tomato seeds accurately predicted germination outcomes. Parameters such as percentage of free space and Sauter diameter were effective in predicting germination. The analysis achieved a high overall prediction potential of 91.9%. However, increased X-ray exposure led to more abnormal seedlings. Silva *et al.* [5] used RGB images of seeds to extract color, morphology and texture features of seeds for the purpose of classification.

In Sri Lanka, tomatoes are cultivated large scale because of their superior nutritional content and flavor [6]. Effective germination and establishment of seeds are crucial characteristics for both field and greenhouse crops. Conducting extensive germination experiments on a large scale is labor-intensive and susceptible to errors made by human observers. This highlights the need for automated approaches [7]. The Seed germination ability among the seed samples is dependent on its phenotypic traits. This paper proposes an autonomous screening approach to automatically detect the phenotypic traits of tomato seeds which can further predict the probability of germination without planting. The proposed autonomous screening approach is a combination of imaging technologies, and machine learning algorithms to provide quick and non-invasive phenotypic analyses of tomato seeds.

## II. METHODOLOGY

Following are the major steps of the autonomous phenotyping approach as shown in Fig. 1.

*Seed Image Acquisition* - The high-resolution camera was used to acquire seed images, which served as the primary data source for subsequent analysis and classification. The dataset consists of 345 images of the resolution 640 x 640.

*Seed Image Preprocessing* - The images were pre-processed by applying filters such as Gaussian blur and median filter, to remove noise and artifacts, which improved the overall quality of the images. Additionally, the contrast of the images was adjusted to ensure consistent illumination across the entire image. Thresholding techniques were applied to separate seeds from the background by defining a pixel intensity threshold. If the seeds had distinct color characteristics, the images were segmented based on color information. Erosion was used to remove small unwanted regions and dilation was used to fill gaps between seed regions.

*Seed Segmentation* - Individual seeds were isolated from the background and other unwanted elements in the images using image segmentation techniques. This step ensured that the machine learning algorithms focused solely on the seeds of interest.

*Seed Image Feature Extraction* - Geometric features were extracted, such as area, perimeter, circularity, and aspect ratio of seed regions. Contours around seed regions were identified and detected to precisely define their boundaries. Connected components in the segmented images were labeled and analyzed using connected component analysis. These features were then used as the key input data for training the machine learning model.

*Seed Sorting* - The trained model is used to predict the seeds which have the probability to germinate in the test set.

### A. Materials

The Department of Agriculture, Sri Lanka (DOA) has so far released 14 different types of tomato varieties and 11 different tomato varieties are currently being produced as certified seeds. Among these varieties “Thilina” is one of the famous tomato varieties among Sri Lankan farmers because of its potential yield is about 40 metric tons per hectare [8]. Over the past 20 years, it has provided higher net economic returns than other varieties. This variety has a semi-indeterminate growth habit (Fig. 2). The tomato seeds for this study were obtained by the Horticultural Crop Research and Development Institute (HORDI), Gannoruwa, Peradeniya.

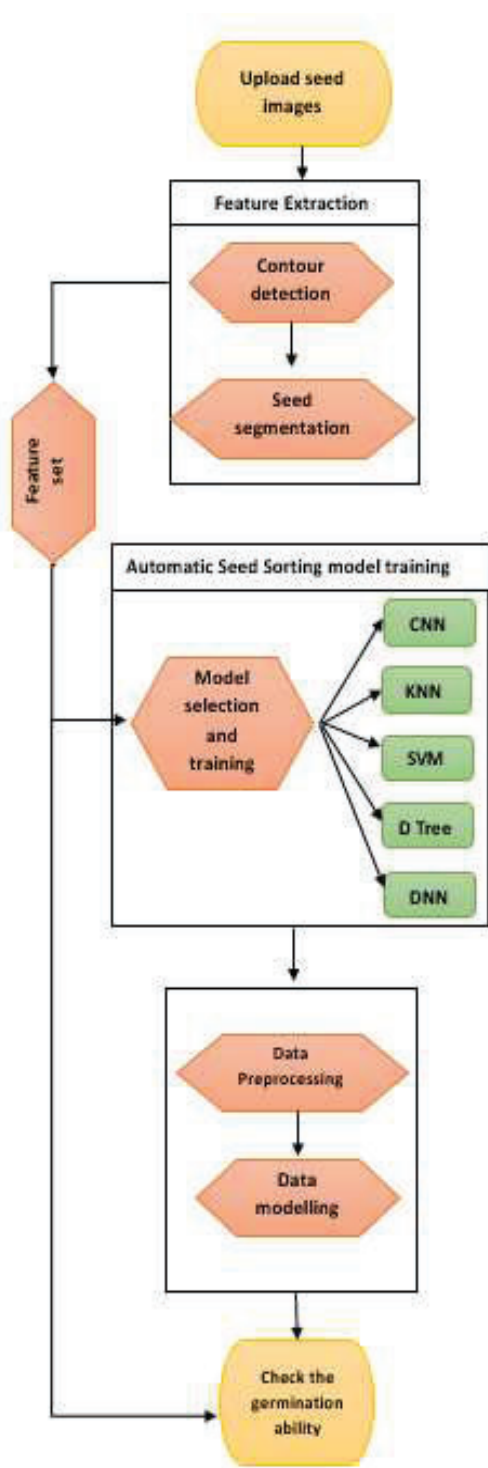


Fig. 1. Graphical representation of the steps for autonomous screening approach



Fig. 2. Tomato commercial variety “Thilina” grown in Sri Lanka. Image is taken from the Department of Agricultural Technology, Faculty of Technology, and University of Colombo

An experimental setup with a Hikvision camera was used for the acquisition of images of the seeds (Fig. 3).



Fig. 3. Image acquisition setup of seeds

### B. Evaluation

A color range was defined in the HSV color space to isolate red-colored objects, representing the tomato seeds (Fig. 4).



Fig. 4. Original image of seeds

Next, a mask was created to separate the seeds from the background, allowing for accurate analysis as shown below. Further, seeds are numbered for future reference (Fig. 5).

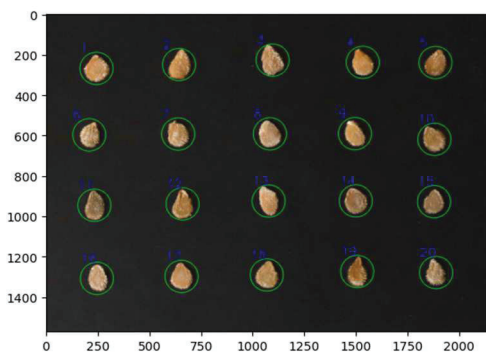


Fig. 5. Seeds masking and labelling

The area determination was crucial for understanding seed distribution and density in agricultural or research contexts.

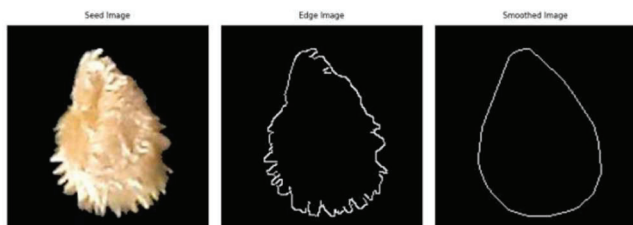


Fig. 6. Edge detection of seeds

Edge detection algorithms like Canny were employed to identify the boundaries of the tomato seeds more precisely (Fig. 6). This step enhanced the accuracy of measurements related to seed shape and size. Additionally, the use of convex hulls aided in outlining the boundaries, providing a clearer representation of seed shapes.

TABLE I. MORPHOLOGICAL TRAITS COLLECTED BY THE SEEDS

| Seed Number | Area   | Perimeter | Circularity | Major Axis Length | Minor Axis Length | Contrast | Dissimilarity | Homogeneity | ASM  | Energy | Correlation | Next Stage(Germinated\Not) |
|-------------|--------|-----------|-------------|-------------------|-------------------|----------|---------------|-------------|------|--------|-------------|----------------------------|
| 1           | 612.72 | 11        | 0.841       | 2.92              | 3.73              | 152.50   | 6.43          | 0.26        | 0.00 | 0.04   | 0.99        | 0                          |
| 2           | 517.31 | 13        | 0.865       | 3.56              | 3.95              | 110.71   | 5.46          | 0.26        | 0.00 | 0.04   | 0.99        | 1                          |
| 3           | 527.02 | 13        | 0.805       | 2.96              | 4.8               | 80.53    | 5.04          | 0.27        | 0.00 | 0.05   | 0.99        | 1                          |
| 4           | 554.56 | 12        | 0.87        | 3.17              | 3.94              | 122.29   | 5.67          | 0.27        | 0.00 | 0.05   | 0.99        | 1                          |

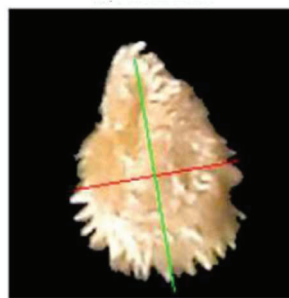


Fig. 7. Markings of major and minor axes

The perimeter and circularity of the detected seed contours were calculated by the functions. These metrics offered insights into the roundness or irregularity of the seeds, which could be indicative of their quality or maturity in agricultural applications. The major and minor axes lengths were measured by fitting an ellipse around the seeds as shown in Fig. 7. These measurements provided information about the overall shape and major axis and minor axis of the seeds, aiding in further characterization.

The feature was collected and saved into a table, including attributes such as number, area (mm<sup>2</sup>), perimeter (mm), circularity, major axis length (mm), minor axis length (mm), contrast, dissimilarity, homogeneity, ASM, energy, and correlation for tomato seeds (Table I). Each row in the table represents a tomato seed, identified by its 'Number' and characterized by its corresponding attributes. For instance, 'Area' denotes the area occupied by the seed in square millimeters, 'Perimeter' represents the perimeter of the seed, 'Circularity' indicates the roundness of the seed, and 'Major Axis Length' and 'Minor Axis Length' specify the lengths of the major and minor axes of the seed. A detailed insight into the morphological and textural characteristics of individual tomato seeds within the analyzed images was provided by this comprehensive data (Table I).

Valuable information for quantitative analysis was presented, enabling comparisons between seeds and a deeper understanding of their variations in size, shape, and texture, which can be crucial in agricultural research, quality assessment, or breeding programs.

### C. Seed Germination Evaluation

After the acquisition of images of seeds, they were transferred to the bubble tray for germination (Fig. 8). The bubble tray was filled with potting mix, leaving a small gap at the top. The potting mixture was ensured to be moist but not waterlogged. Tomato seeds were planted at a depth of 1/4 to 1/2 inch in the bubble cell. Each cell was labeled with a pre-defined seed number. The seeds were watered gently using a spray bottle. The potting mixture was kept consistently moist but not overly wet.

The bubble tray was placed in a warm location with temperatures around 21-27°C and adequate light for seed germination. The tray was monitored for signs of germination, which usually takes around 5-12 days.



Fig. 8. Seeds were germinated in the tray

The whole seed data set was split into training and testing. Corresponding to the training seeds, (Table I) was manually annotated by adding the *nextStage* Column by looking at the germination of the training seeds.

Various classifiers including CNN (Convolution Neural Network), Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Decision Trees, and a Deep Neural Network (DNN) were trained subsequently. The training set was used to train each model, and then the test set was used to evaluate the performance of each model, with classification reports produced to showcase their individual performance.

The CNN model summary is displayed in Table II.

TABLE II. CNN MODEL SUMMARY

| Layer (type)    | Output Shape        | Parameters |
|-----------------|---------------------|------------|
| conv2d_8        | (None, 62, 126, 32) | 896        |
| max_pooling2d_8 | (None, 31, 63, 32)  | 0          |
| flatten_8       | (None, 62496)       | 0          |
| dense_16        | (None, 128)         | 7999616    |
| dropout_1       | (None, 128)         | 0          |
| dense_17        | (None, 3)           | 387        |

### III. RESULTS AND DISCUSSION

A systematic approach to model selection was provided by evaluating multiple algorithms on the dataset and aiding in identifying the most effective model for predicting seed stages based on the given features. The best-performing model, determined by the highest accuracy, was saved for potential future use. The performance scores of each model are given in Table II.

$$accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$precision = \frac{TP}{TP+FP} \quad (2)$$

$$recall = \frac{TP}{TP+FN} \quad (3)$$

$$F1 - score = \frac{2 \times precision \times recall}{precision+recall} \quad (4)$$

TABLE III. PERFORMANCE SCORES OF EACH MODEL

| Classifier | Precision | Recall | F1-score | Accuracy |
|------------|-----------|--------|----------|----------|
| CNN        | 0.87      | 0.93   | 0.9      | 0.86     |
| SVM        | 0.8       | 0.56   | 0.64     | 0.77     |
| KNN        | 0.99      | 0.88   | 0.93     | 0.94     |
| D Tree     | 0.86      | 0.77   | 0.81     | 0.83     |
| DNN        | 0.63      | 0.35   | 0.39     | 0.74     |

The accuracy refers to the percentage of correct predictions made by the model when compared with the actual classifications in the test data as shown in the equation (1). The precision as shown in the equation (2) was described as the proportion of positive predictions recall was defined as the proportion of actual positives that were correctly identified as positive as shown in equation (3). F1-score was calculated as the harmonic mean of precision and recall as shown in equation (4). Accuracy was defined as the overall proportion of correct predictions. The best result, with an accuracy of 0.94, was achieved by KNN. Slightly lower accuracies of 0.86 and 0.77 were obtained by CNN and SVM, respectively. Decision Tree and DNN had the lowest accuracies of 0.83 and 0.74, respectively. It was concluded that KNN was the best classifier for this task since it had the highest accuracy and was also competitive in terms of precision, recall, and F1-score. Once the model was trained, a new data point from the extracted feature was created and passed to the model for the prediction.

### IV. CONCLUSION

This study addresses the challenges posed by the lack of a reliable and adaptable prediction approach for tomato seed germination ability. To overcome this challenge, this paper proposes an automated system that uses image analysis and machine learning techniques to automatically detect the phenotypic traits of tomato seeds which can further predict the probability of germination without planting. The phenotypic characteristics of tomato seeds were extracted using sophisticated feature extraction methods. The study employed machine learning-based approaches, and features were passed through generalized machine learning models such as KNN, SVM, Decision Tree, CNN, or DNN, depending on the approach. The best result, with an accuracy of 0.94, was achieved by KNN. It was concluded that KNN was the best classifier for this task since it had the highest accuracy and was also competitive in terms of precision, recall, and F1-score.

To the best of our knowledge, this study is the first in Sri Lanka which automated conventional seed tests by means of computer vision and machine learning for a local Tomato variety called “Thilina”. Further, this research can be extended to predict the germination ability of tomatoes and other cultivars locally and globally.

In future, the proposed system will be used as an iterative pipeline, to improve the whole life cycle of the tomato plant. The hardware setup will also designed and developed to automatically capture the images throughout the growth stage, extract phenotypic characteristics, and process them subsequently.

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